Heart-Disease Risk Prediction: A Leakage-Free, Calibrated and Fairness-Audited Re-implementation of Shi, Bechler & Ho

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Course: UE24CS352A | Repository: https://github.com/coderRoopa/ML_mini_project

1. Problem statement

Shi et al. predict myocardial-infarction risk from routine outpatient data (logistic regression: AUROC 0.86-0.87, recall 0.94). Their conclusions rest on **one random 60/20/20 split** (test n of about 61), model choice on a tiny validation set, an SVM that collapsed to AUROC 0.50, a fixed 0.5 threshold, no calibration check and no subgroup analysis, even though sex disparity in cardiac care is their motivation. **Our problem:** build a binary classifier that flags high-risk patients from 13 routine features with high sensitivity, and estimate *honestly* (with uncertainty) its discrimination, calibration, operating point, sex-specific performance and interpretability.

2. Dataset

Kaggle *Heart Attack Analysis & Prediction Dataset* (derived from UCI Cleveland): 303 patients, 13 features (age, sex, chest-pain type *cp*, resting BP, cholesterol, fasting sugar, rest ECG, max heart rate, exercise angina, ST depression *oldpeak*, ST slope, vessels *caa*, thallium *thall*) and a binary outcome. **Data checks the paper did not report:** (i) 1 duplicate removed, leaving **302 patients (96 women, 206 men)**; (ii) invalid codes *caa*=4 and *thall*=0 treated as missing and imputed inside the pipeline; (iii) **label direction:** in the file, *output*=1 has higher max heart rate and *less* exercise angina, ST depression and diseased vessels, i.e. it marks the *no-disease* group. Our automatic check therefore defines **risk = 1 - output** (46% prevalence; majority-class accuracy 0.543). The outcome is angiographic coronary disease, not a documented MI.

3. Approach

- **Leakage-free preprocessing** in every *Pipeline* (re-fit per training fold): median / most-frequent imputation, one-hot encoding of the nominal *cp, restecg, slp, thall*, standardisation.
- **Nested repeated stratified CV** for 9 model families (logistic regression, elastic net, SVM linear/RBF/poly, MLP, bagging, random forest, gradient boosting) plus a majority baseline: outer 5-fold x 3 repeats = 15 out-of-sample estimates (mean ± SD); inner 3-fold randomized search on AUROC.
- **Model choice:** Nadeau-Bengio corrected paired t-test with Holm adjustment against the best model, then the **one-standard-error rule** (simplest model statistically indistinguishable from the best).
- **Calibration and honest threshold:** sigmoid recalibration; the decision threshold for a 95% sensitivity target is chosen on *inner* out-of-fold predictions of each training fold and applied unchanged to the held-out fold (**nested**), and contrasted with the usual naive choice on the evaluated data.
- **Audit and interpretation:** per-sex metrics and male-minus-female gaps with bootstrap CIs; odds ratios with bootstrap CIs; permutation importance on held-out folds.
- **Unsupervised analysis:** PCA, k-means, DBSCAN on scaled/encoded data, with settings chosen by silhouette (labels never used for selection); the paper's raw-data setup is replicated for comparison.

4. Implementation overview

Python 3 and scikit-learn, organised as a tested repository: `src/` (*data, preprocessing, models, evaluation, calibration, fairness, interpret, clustering, plots, pipeline*), `main.py` (full pipeline CLI, `--fast` for a smoke test), `predict.py` (scores new patients with the saved calibrated model and its threshold), `notebooks/`, `tests/` (pytest) and `outputs/` (figures, CSV tables, model, summary). Every number below is generated by code (seed 42) and saved in `outputs/`.

5. Results

Model (nested CV, 15 folds)	AUROC	Accuracy	Recall	F1	Brier
Majority baseline	0.500 ± 0.000	0.543 ± 0.006	0.000	0.000	0.457
Logistic regression (selected)	**0.911 ± 0.040**	**0.845 ± 0.044**	**0.777 ± 0.094**	**0.819 ± 0.056**	**0.123**
Elastic net	0.914 ± 0.039	0.850 ± 0.049	0.777 ± 0.094	0.823 ± 0.062	0.118
SVM linear / RBF / poly	0.911 / 0.908 / 0.901	0.839 / 0.844 / 0.776	0.775 / 0.789 / 0.780	0.812 / 0.821 / 0.766	0.120 / 0.120 / 0.159
MLP	0.875 ± 0.049	0.791 ± 0.037	0.765 ± 0.093	0.769 ± 0.045	0.163
Bagging / RF / GradBoost	0.912 / 0.916 / 0.903	0.828 / 0.836 / 0.828	0.755 / 0.782 / 0.762	0.797 / 0.812 / 0.799	0.124 / 0.128 / 0.129

Table 1. Out-of-sample performance (threshold 0.5 on predicted probability). Full per-model table: outputs/tables/cv_summary.csv.

- **Model differences are noise.** All families reach AUROC 0.875-0.916; after Holm correction none differs significantly from the best (random forest, 0.916); the smallest raw p is 0.035 (SVM poly, Holm 0.28). The one-SE rule therefore selects the simplest model, **logistic regression**, which is also the most interpretable.
- **A single split is unreliable.** Repeating the paper's protocol with 200 seeds, test AUROC of logistic regression ranges **0.82-0.98** (random forest 0.79-0.97). The paper's SVM-RBF collapse (AUROC 0.50) does not recur once the model is pipelined and tuned (AUROC 0.908).
- **Calibration** is already good (slope 1.12; Brier 0.122, 0.121 after sigmoid recalibration), so recalibration changes little.

Operating point	Sensitivity [95% CI]	Specificity [95% CI]	Precision
Threshold 0.50	0.775 [0.71, 0.84]	0.884 [0.83, 0.93]	0.849
Naive threshold 0.19 (optimistic)	0.964 [0.93, 0.99]	0.567 [0.49, 0.64]	0.652
Nested threshold, median 0.21 (honest)	**0.942 [0.90, 0.98]**	**0.591 [0.52, 0.67]**	**0.660**

Table 2. Operating points for a 95% sensitivity target (pooled held-out predictions).

- **The sensitivity target costs specificity.** The honest nested threshold reaches sensitivity 0.94 (just under the 0.95 target) at specificity 0.59, versus 0.88 at threshold 0.5. The naive threshold overstates sensitivity (0.964) because it is tuned on the data it is evaluated on.

- **Sex audit (nested threshold).** AUROC: women 0.931 [0.86, 0.98], men 0.880 [0.83, 0.92]. Sensitivity: 0.875 vs 0.956 (gap +0.08, CI [-0.05, 0.23], not significant). **Specificity: 0.708 vs 0.500 (gap -0.21, CI [-0.35, -0.07], significant)**: at a shared threshold men receive far more false alarms. With only 96 women, intervals are wide.

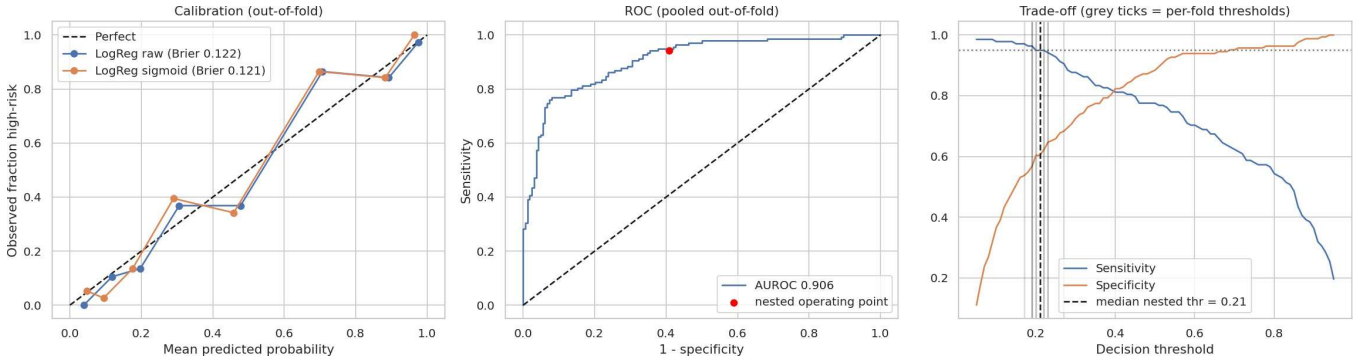


Figure 1. Out-of-fold calibration curves, pooled ROC (AUROC 0.906, red = nested operating point) and sensitivity/specificity versus threshold (grey = per-fold thresholds).

- **Interpretation** (risk = disease). Odds ratios per 1 SD / category: vessels *caa* 1.82 [1.53, 2.14], asymptomatic chest pain 1.51, reversible thallium defect 1.44, ST depression 1.42, exercise angina 1.38, male sex 1.32; protective: normal thallium 0.60, higher max heart rate 0.72, atypical angina 0.72, up-sloping ST 0.75. Held-out permutation importance agrees: *caa* dominates (AUROC drop 0.061), then *thall* and *cp*.

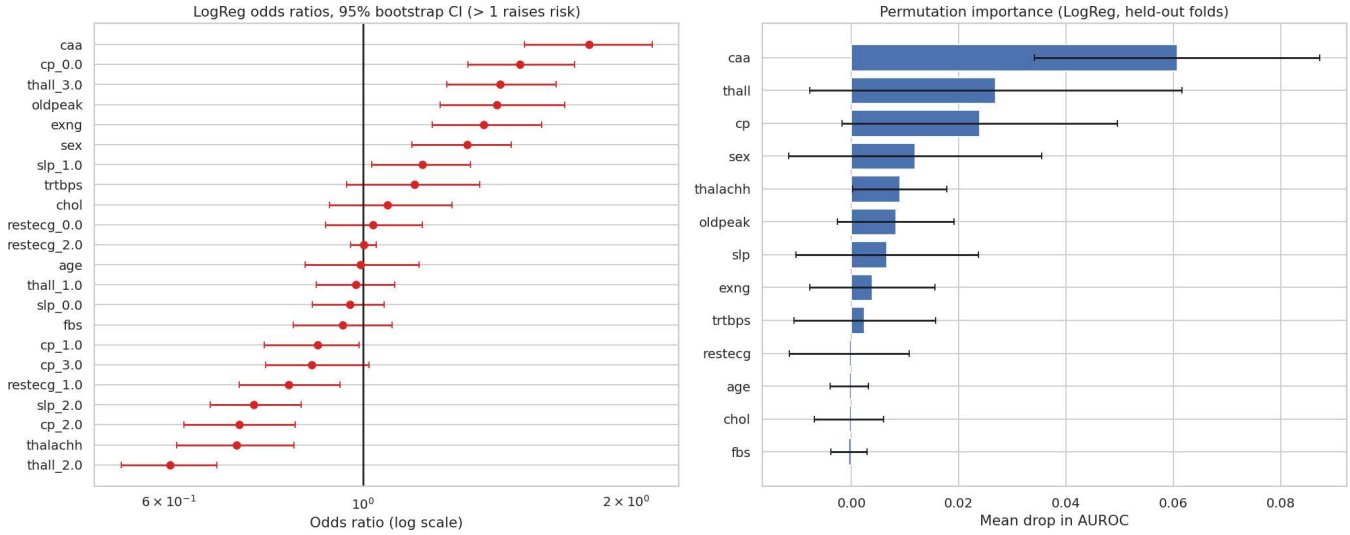


Figure 2. Left: logistic-regression odds ratios with 95% bootstrap CIs (OR above 1 raises risk of disease). Right: permutation importance (drop in AUROC) on held-out folds; error bars = SD across folds.

- **Unsupervised structure is modest.** On raw data the paper's setup gives ARI 0.02 (as reported); scaling and encoding raises k-means to **ARI 0.30** (NMI 0.23, silhouette 0.41; 2 PCs, k=2 chosen without labels). The clusters are clinically meaningful (age 60, ST depression 1.7, exercise angina 53%, 75% disease vs age 50, 0.5, 15%, 20%). DBSCAN finds no useful density structure (best ARI 0.08 with 24% noise).

6. Conclusions

- With about 300 patients, **evaluation design matters more than model choice**: nine model families are statistically indistinguishable (AUROC about 0.91), and one 61-patient split can swing AUROC by 0.16.
- A simple, calibrated **logistic regression** is sufficient, transparent and well calibrated; the clinically sensible drivers (vessels, thallium, chest pain, ST depression, exercise angina) emerge from both odds ratios and permutation importance.
- A 95% sensitivity target is achievable honestly (0.94) but costs about 30 points of specificity, and the **cost falls unevenly on men** (specificity 0.50 vs 0.71 for women), an audit absent from the original paper.
- AUROC is comparable to the paper (0.91 vs 0.86-0.91) but recall and precision are not: our label check suggests the paper's positive class (*output*=1) is the no-disease group, so we compare only the symmetric AUROC.
- **Limitations:** single-centre, 302 patients, outcome is angiographic disease rather than documented MI, no external validation, wide subgroup CIs. **Future work:** validate on other UCI cohorts (Hungarian, Swiss, Long Beach), add richer EHR variables, and evaluate sex-specific thresholds.

References. *Shi K., Bechler K., Ho V. Predicting myocardial infarction risk using supervised and unsupervised machine learning methods (2021). Nadeau C., Bengio Y. Inference for the generalization error. Machine Learning 52 (2003). Detrano R. et al. Am J Cardiol 64 (1989) (Cleveland data). Rahman R. Heart Attack Analysis & Prediction Dataset, Kaggle.*